

Standard Operating Procedure ***Bench Grinder***

Snap-on Bench Grinder

A bench grinder is an essential shop tool used for sharpening cutting tools, deburring metal parts, and cleaning surfaces with wire wheels. It utilizes two high-speed abrasive wheels mounted on a motor mandrel. Because these wheels operate at high rotational velocities and can shatter (explode) if damaged, strict adherence to setup clearances and wheel integrity is mandatory.

Potential Hazards

- **Mechanical**: High-speed wheel explosion/fragmentation, entanglement in the spindle, and pinch points at the tool rest.
- **Operational**: High-velocity sparks, intense heat of the workpiece, and metallic dust.
- **Respiratory**: Inhalation of abrasive grit and metal particulates.
- **Energy Sources**: High-speed rotational kinetic energy and 120V electrical power.

Personal Protective Equipment (PPE)

PPE Item	Requirement
Eye Protection	**ANSI Z87.1-compliant safety goggles:** Mandatory. Standard glasses are insufficient against high-velocity wheel fragments.
Face Protection	**Face Shield:** Highly recommended in addition to goggles to protect against "wheel burst" and heavy sparks.
No Gloves Advised	**Strictly Forbidden:** Gloves are an extreme entanglement hazard. If a glove snags on the wheel, it will pull the hand into the tool rest.
Hearing Protection	Should be worn during prolonged grinding operations.



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PPE Item	Requirement
Apparel & Hair	Tie back long hair and secure loose clothing. Remove all jewelry, especially rings and watches.
Footwear	Appropriate enclosed footwear must be worn at all times.

Before Use

- **The "Ring Test"**: Prior to mounting any new wheel, suspend it by the center hole and tap it with a non-metallic object. A clear metallic "ring" indicates a sound wheel; a "thud" indicates a crack, and the wheel must be discarded.
- **Speed Rating Verification**: Ensure the Maximum Operating Speed (RPM) of the abrasive wheel is **equal to or greater than** the rated RPM of the grinder motor.
- **Tool Rest Gap**: Adjust the tool rest to within **1/8 inch** (3.2 mm) of the wheel surface. This prevents the workpiece from being pulled down between the wheel and the rest.
- **Spark Guard (Tongue Guard)**: Adjust the spark deflector at the top of the wheel opening to within **¼ inch** (6.4 mm) of the wheel.
- **Guard Integrity**: Ensure all side guards and eye shields are securely in place and clean for visibility.

During Use

- **Startup Safety**: Stand to one side of the wheel when turning on the grinder. Allow it to run at full speed for at least one minute before applying work. If the wheel is going to burst, it will typically do so upon startup.
- **Grinding Zone**: **Never** grind on the side of a standard wheel. Use only the face (periphery) of the wheel unless it is specifically designed for side-pressure.



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- **Work holding**: Use the tool rest to support the workpiece. Hold the part firmly but do not "jam" it into the wheel.
- **Cooling**: Periodically dip the workpiece in a water quench pot to prevent overheating and loss of temper in the metal.
- **Vibration**: If the machine vibrates excessively, shut it off immediately. The wheel may be out of balance or cracked.

After Use

- **Coast-Down**: Shut off the power and wait for the wheels to come to a complete stop on their own. **Never** use an object or your hand to stop the wheel.
- **Wheel Dressing**: If the wheel is "loaded" (clogged with metal) or out of round, use a diamond or star-wheel dresser to restore the face.
- **Cleanup**: Use a brush (never hands) to clear metal dust and grit from the base and surrounding table.
- **Floor Safety**: Sweep the area. Abrasive dust is slippery and can be hazardous if tracked through the shop.

Emergency Equipment and Procedures

- **Incident Reporting**: Report all wheel failures, "near-misses," and injuries to the Laboratory Project Supervisor immediately.
- **Emergency Stop**: Know the location of the main power switch. In the event of a wheel burst, disconnect power immediately if safe to do so.
- **First Aid**: Know the location of the nearest eye wash station, as fine grit can bypass even well-fitted goggles.